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碩士論文計畫書
MASTER'S THESIS PROPOSAL

隱形帳簿：量化印尼平台交易額與平台收入之差距
The Invisible Ledger: Quantifying Indonesia's Platform Transaction-Revenue Gap

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1. Research Background and Motivation

Digital commercial activity has expanded on a scale that is easy to observe but surprisingly difficult to translate into one economic measure. Commerce can appear as activity facilitated through platforms, revenue recognized by platform firms, sales recorded by businesses, national e-commerce estimates, payment-system transactions, or information available to public institutions. These quantities are related to the same broader transformation, but they are not the same economic object and need not move together.

That distinction becomes economically important when the measures do not lead to the same account of change. When the digital economy grows, what exactly is growing, and where does that growth actually show up? Transaction activity, company performance, business participation, and broader economic measures capture different parts of that process and can move differently. A narrow focus on any one record can therefore misstate where digital transformation is occurring or how its scale should be interpreted. This measurement problem is particularly relevant to digitally mediated activity that crosses organizational boundaries, where economic activity may be distributed across platforms, independent participants, payment systems, and other records rather than contained within a single firm.

This tension motivates the central question of the thesis: If digital commercial activity has become so large, where does that growth actually show up in the economy, and why can its apparent impact differ depending on how it is measured?

The platform transaction-revenue boundary provides a natural place to begin. Multisided-platform theory explains why platforms may coordinate large volumes of third-party commerce while recognizing commissions, fees, advertising, logistics, travel, financial-service, or other revenue within a narrower accounting perimeter (Rochet & Tirole, 2003; Armstrong, 2006; Hagiu & Wright, 2015). The existence of a gap is therefore not itself anomalous. The more informative question is whether transaction activity and platform revenue provide the same longitudinal account of growth and, when they diverge, what mechanisms explain the difference.

That longitudinal comparison provides the first empirical step. Once the company-level relationship is understood, a further question follows: where does the broader digital transformation appear outside issuer accounts, and when do the resulting digital traces become useful to statistical or administrative institutions?

Following that sequence requires a setting in which several independent records can be observed without treating them as interchangeable measures. Indonesia provides such a setting through issuer disclosures, official BPS e-commerce evidence, Bank Indonesia payment statistics, and an evolving platform-reporting architecture. The thesis therefore follows digital-commercial activity across these records to determine where the economic interpretation changes and what can be learned by reconciling those changes.

2. Research Questions and Objectives

The analysis follows this sequence through three linked research questions. RQ1 examines the firm-level boundary between facilitated activity and recognized revenue. RQ2 asks what the firm-level evidence leaves out by moving to business participation, sales channels, and payments. RQ3 then examines the institutional boundary between the existence of digital traces and their eventual statistical or administrative use.

Research question	Purpose	Primary evidence
RQ1. How does growth in digitally mediated commercial activity translate into recognized platform-company revenue, and what mechanisms explain divergence?	Test whether transaction activity and revenue give the same longitudinal growth signal; reconcile incentives, monetization, accounting basis, and perimeter.	Issuer annual transitions; revenue-component reconciliations.
RQ2. Where else does digital-commercial growth appear in business participation, sales channels, and payment activity?	Determine how the broader economic story changes outside issuer accounts.	BPS national/channel/business evidence; Bank Indonesia payments as independent support.
RQ3. Under what conditions do digital traces become statistically or administratively usable information?	Separate private traceability, legal reportability, operational linkage, and observed institutional use.	PMK 37/2025, DJP materials, BPS/BI institutional evidence.

Together, the three questions aim to identify when the observational record changes the economic conclusion, explain material disagreements where the sources allow, locate broader digital growth in participation and channel structure, and distinguish private traceability from institutionally usable information.

3. Literature Review and Research Gap

3.1 Platform economics and mediated activity

Platform economics provides the theoretical foundation. Two-sided and multi-sided platforms coordinate interactions across participant groups and choose price structures across those groups, so facilitated transaction activity and platform revenue need not move proportionally (Rochet & Tirole, 2003; Parker & Van Alstyne, 2005; Armstrong, 2006). Hagiu and Wright (2015) further distinguish multi-sided platforms from reseller or vertically integrated models by the control retained by affiliated participants. This literature establishes why transaction activity and revenue can diverge; the next question is how the underlying activity is translated into the accounting numbers that investors and researchers actually observe.

3.2 Revenue recognition, incentives, and comparability

Accounting rules and business-model design determine how underlying commerce becomes corporate revenue. IFRS 15 distinguishes principal and agent relationships and therefore affects gross-versus-net revenue presentation. Customer incentives, advertising, logistics, subscriptions, travel, financial services, acquisitions, and reporting perimeter can all change revenue without equivalent changes in facilitated commerce. Financial-statement comparability matters because users must understand how similar economic events map into reported numbers; De Franco, Kothari, and Verdi (2011) show that comparability is associated with a stronger analyst information environment. Yet even a well-reconciled issuer account describes only the company perimeter, which leaves open where digital growth appears in the broader economy.

3.3 Measuring the digital economy

Digital-economy measurement literature addresses that broader boundary. Official frameworks distinguish the underlying buyer-seller transaction from the intermediation service provided by the platform, while national-accounting work rejects the shortcut of treating gross transaction value as value added or an alternative GDP measure. Digitalization can nevertheless create measurement difficulties involving classification, business perimeter, cross-border activity,

prices, private-source data, and publication revision (Ahmad & Schreyer, 2016; IMF, 2018; OECD, 2023). These frameworks clarify what aggregate measures mean, but they still say little about how participating businesses differ in their own records and payment practices.

3.4 Business records, digital footprints, and third-party information

Research on small firms and informality therefore adds the participant side of the picture. Business practices and conventional recordkeeping vary substantially (La Porta & Shleifer, 2014; Ulyssea, 2018; McKenzie & Woodruff, 2017), while digital footprints can contain economically useful information even where conventional records are incomplete (Berg et al., 2020). A related public-finance literature shows that third-party information and transaction paper trails can matter for verification when records are reported and linked to the relevant taxpayer or transaction (Kleven et al., 2011; Pomeranz, 2015; Slemrod, 2019). This moves the problem from whether a digital trace exists to whether it is structured, linked, and institutionally usable.

3.5 Research gap

Taken together, these literatures explain the individual boundaries that the thesis encounters: platform economics explains why facilitated activity and revenue differ; accounting explains how activity enters the issuer perimeter; digital-economy measurement distinguishes gross transactions from value added; business research describes variation in participant records; and third-party-information research explains why linkage matters administratively. What remains less developed in the present setting is a source-auditable account that follows the same digital-commercial transformation across those boundaries and asks when the resulting economic interpretation changes. The thesis addresses that gap by tracing where digital growth becomes visible, explaining why selected records diverge, and identifying what additional information is recovered through reconciliation.

4. Conceptual Framework: Economic Translation and the Invisible Ledger

The literature therefore suggests a sequence rather than a set of competing ledgers. A digital commercial event may appear first as facilitated activity, then as a corporate accounting entry, a merchant record, a payment-system trace, a statistical observation, or an administrative record. These representations are not expected to be identical because each is produced for a different purpose. The analytical question is whether using one in place of another changes the economic conclusion.

A cross-record difference matters when one measure is being used to stand in for a broader process, another legitimate measure points to a materially different interpretation, and understanding the difference requires reconciling scope or mechanism. The thesis therefore treats disagreement as informative only when it affects a substantive conclusion about growth, participation, channel structure, or institutional visibility.

This is where the term Invisible Ledger becomes useful. It refers to the fuller economic account that becomes recoverable when the relationships, disagreements, and boundaries among partial records are reconciled.

The transaction-revenue relationship provides the first empirical clue, but the concept extends across the full sequence of issuer, business, payment, statistical, and administrative records. Although the empirical analysis remains focused on Indonesian platform commerce, the same measurement logic can arise in adjacent digitally mediated activity, including gig work and creator or influencer commerce, where economic activity crosses organizational boundaries and no single corporate, participant, payment, or statistical record captures the complete process. The analytical objective is to determine what each record captures, where their interpretations diverge, and what becomes visible through reconciliation.

5. Data and Research Design

Because each stage of the research question observes a different unit, the empirical design remains modular. The modules are linked by the sequence of the argument, but their observations are never added into one sample N or treated as if they measured the same object.

Module	Unit / coverage	Role	Key boundary
Indonesia issuer	13 direct candidate periods; 9 direct-candidate transitions	Main longitudinal test of RQ1	Final Blibli/Bukalapak scope remains advisor-dependent.
Mechanism	Issuer-period components where disclosed	Explain selected divergences	Arithmetic reconciliation is not causal decomposition.
BPS national/business	2022-2024 national indicators; published business-level 2023 comparison	RQ2: growth anatomy, channels, participation, recordkeeping	Different unit from issuer data; business-level associations are observational.
Bank Indonesia	SPIP monthly/annual histories; QRIS reports	Independent traceability support	Payments are not e-commerce sales.
Institutional	PMK 37/2025 and DJP implementation documents	RQ3: legal vs operational visibility	Legal architecture does not prove operational linkage or effects.
ASEAN/global	Six separate ASEAN histories; 48 global issuer-years	External corroboration / robustness	Not pooled with Indonesia; purposive and heterogeneous.

5.1 Issuer admission rules

An issuer-period is eligible only when transaction and revenue measures cover the same period; geography and business scope can be evaluated; currency and units are known; original, revised, and pro-forma bases are labelled; definitions are recorded; derived values have transparent source inputs; and material structural breaks are flagged. Annual totals, quarters, publication vintages, and source inputs are not treated as independent observations.

The direct candidate inventory contains 13 economic periods: seven Blibli 3P periods (FY2019-FY2025), four Bukalapak Group periods (FY2020-FY2023), and two Tokopedia e-commerce periods (FY2022-FY2023). Tokopedia FY2021 is excluded for period mismatch, and Bukalapak FY2024 is excluded because transaction value and revenue cover different durations. Grab and Shopee country reconstructions remain conditional sensitivity evidence rather than direct core observations.

6. Preliminary Evidence and Feasibility

The modules already yield preliminary evidence along the same sequence as the research questions. The results first establish that issuer growth signals can diverge, then examine why, and finally move outward to broader business, payment, and institutional evidence. The final main-sample boundary remains subject to advisor approval.

6.1 Issuer growth signals can diverge

Across nine direct-candidate transitions, revenue grows faster in six and transaction activity grows faster in three; three transitions have opposite signs. The median absolute transaction/revenue growth divergence is 42.94 percentage points. This is a diagnostic of the

candidate evidence, not a population parameter. The sign reversals make the next question one of mechanism rather than simply magnitude: what changed inside the revenue measure when the underlying activity moved differently?

The cleanest mechanism case is Tokopedia FY2022-FY2023: transaction activity falls 8.90% while selected third-party net segment revenue rises 53.20%. Approximately 60.56% of the arithmetic increase in net revenue is associated with lower customer incentives and 39.44% with higher gross revenue. The reconciliation shows how recognized revenue can strengthen even as facilitated activity weakens. But issuer accounts cannot show whether broader digital growth is appearing elsewhere in the economy, so the analysis then turns to official BPS evidence.

6.2 Broader e-commerce growth has a different composition

BPS broadens the lens from the platform firm to the national e-commerce population. Total nominal e-commerce transaction value rises from Rp1,100.87 trillion in 2023 to Rp1,288.93 trillion in 2024 (+17.08%), while estimated e-commerce businesses rise from 3,816,750 to 4,400,972 (+15.31%). Implied nominal value per estimated business therefore rises only 1.54%. This arithmetic decomposition does not identify firm entry or productivity, but it shows that recent aggregate growth cannot be interpreted only as larger activity per business.

The channel split then changes the picture again. Marketplace value rises only 1.45%, while the non-marketplace component rises 20.57%; arithmetically, the non-marketplace component accounts for about 98.46% of the nominal increase in the published total. The decomposition therefore places almost all of the measured nominal increase outside the marketplace component. It does not identify why channel growth rates differ, which motivates the next question: whether marketplace and non-marketplace participants differ in observable recordkeeping and payment practices.

6.3 Channel participation is associated with different recordkeeping

BPS's published business-level analysis for reference year 2023 provides that participant-side comparison. Among 6,148 marketplace users and 33,588 non-marketplace users, complete financial statements are reported for 28.63% versus 12.25%, while cash use is 37.74% versus 83.29%. BPS's adjusted model also reports a positive association between financial-statement ownership and marketplace use. The evidence remains observational rather than causal, but it suggests that the channel with stronger conventional recordkeeping is not necessarily the channel where the latest measured growth is concentrated. That tension makes payment-system evidence a useful next check rather than an alternative measure of the same commerce.

6.4 Payment traces and administrative records add separate views

Bank Indonesia provides that independent payment-system view. Between 2023 and 2024, BPS e-commerce value grows 17.08%, while electronic-money shopping, mobile-banking payment/purchase, and QRIS transaction values grow at substantially different rates. The contrast is expected because payments and sales are different economic objects. Its value is therefore not to validate BPS or issuer data, but to show that abundant digital traces can expand on a different trajectory from the commerce they help record.

Once trace abundance is separated from economic meaning, the remaining question is institutional: when can those traces actually be used by public administration? PMK 37/2025 shows a legal bridge built around seller identity, transaction-linked turnover, withholding, and reporting rather than platform corporate revenue. Current evidence establishes that legal architecture, but the implementation postponement prevents claims about successful operational matching, compliance, or revenue effects.

7. Expected Contribution, Stakes, and Limitations

7.1 Expected contribution

Read together, these stages produce four connected contributions. First, the thesis provides a source-auditable account of where digital-commercial growth appears across different records. Second, it explains selected disagreements through mechanisms such as monetization, incentives, accounting presentation, business perimeter, participation, and channel composition. Third, it shows that disagreement among records can reveal economic structure rather than merely constitute measurement noise. Fourth, it distinguishes the existence of digital traces from the additional conditions required for statistical or administrative use. More broadly, the framework provides a way to study digitally mediated economic activity whose observable footprint is distributed across multiple partial records rather than contained within a single organizational account.

7.2 Why the problem matters

The problem matters because different records can lead to different diagnoses of the same transformation. Revenue growth can imply a different platform-performance story from transaction activity; marketplace-centered evidence can mislocate where national e-commerce growth occurs; aggregate value can obscure whether expansion reflects broader participation; and payment growth can be mistaken for equivalent commerce growth. At the institutional level, rich private traces still require identity, reporting rules, transmission, and matching before they become operationally useful. The concern is therefore not that the records differ, but that one may be used as a proxy for another without reconciling the boundary.

7.3 Scope and limitations

The design has five principal limitations. First, the issuer sample is small and selected by the availability and comparability of disclosures. Second, the BPS business-level relationships are observational. Third, Bank Indonesia payment measures provide a separate view of digital trace activity rather than a measure of e-commerce sales. Fourth, PMK 37/2025 establishes legal reporting architecture, while postponed implementation limits inference about operational linkage or policy effects. Fifth, the ASEAN and global issuer modules are purposive external corroboration rather than representative regional or global samples.

8. Advisor Decision Gates and Work Plan

The empirical foundation is sufficiently developed for the proposal, while the remaining advisor decisions are narrow and explicit: whether Blibli's travel-inclusive 3P segment is acceptable, whether Bukalapak's overseas scope is acceptable, and which resulting issuer boundary should become the final longitudinal main sample. Grab and Shopee remain conditional sensitivity cases unless the advisor requests otherwise. Once those decisions are fixed, the full manuscript can be completed without changing the broader research sequence.

Period	Planned work
September 2026	Advisor review of proposal and empirical package; oral proposal examination; freeze final issuer-sample rule and required checks.
October-November 2026	Complete requested mechanism reconciliations and source concordance; incorporate committee feedback; freeze thesis tables/figures.
Before final degree defense	Complete the full manuscript under the approved boundary; retain supporting payment/ASEAN/global/institutional modules only where they advance the final argument.

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